

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Agroecosystems: abiotic & biotic factors - Set 1

Mark scheme

AQA spec 3.5.1.1; Biological resources; textbook pp. 306-320

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 3 points below. 1 B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings 1 B is warmer / longer thermal growing season because south-facing slopes receive more insolation – valley floors are frost pockets where cold dense air collects 1 But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • insolation → incoming solar radiation / energy arriving from the Sun	3

Question	Marking guidance	Marks
01.2	Any 3 of 4 1 mark each for any 3 of the 4 points below, in any order, maximum 3. 1 Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) 1 Woven plastic cloches / fleece over rows to retain warm air near the ground 1 Burning oil / gas or using large fans in orchards to prevent frost damage to blossom 1 Site selection (aspect, avoiding frost pockets) – heated buildings and shelter for livestock Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.3	1 mark each 1 mark for each of the 2 points below. 1 Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) 1 Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	1 mark each 1 mark for each of the 2 points below. 1 Long days increase milk production and egg laying – poultry grow best with short days – sheep mate as days shorten in autumn 1 Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	
02.1	1 mark each 1 mark for each of the 3 points below. 1 Plants close their stomata to reduce transpiration losses 1 Closed stomata stop the absorption of carbon dioxide so photosynthesis and growth stop 1 Less water for nutrient absorption / transport and cell turgidity – severe shortage dehydrates cells and inhibits biochemical reactions Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. transpiration → water lost as vapour from the leaves	3

Question	Marking guidance	Marks
02.2	Marks shared Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. up to 2 Anaerobic soil favours denitrifying bacteria and inhibits nitrifying bacteria so nitrate is lost faster than it forms – higher risk of fungal disease – root respiration inhibited up to 2 Drainage ditches or pipes up to 2 Deep ploughing – avoiding compaction by machinery / livestock – adding organic matter to encourage worms Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • waterlogged → soil pores filled with water so there is no air • denitrifying → nitrate converted back to nitrogen gas • nitrifying → ammonium oxidised to nitrite and then nitrate	4
02.3	1 mark each 1 mark for each of the 2 points below. 1 Groundwater often contains dissolved salts – when the water evaporates from the soil the salts are left behind and accumulate 1 High soil salinity causes osmotic dehydration of roots / crop death – washing out with extra water uses more water and salinises rivers Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • salinisation → salt building up in the soil / soil becoming salty	2
02.4	1 mark for the point below. 1 Add crushed lime (calcium carbonate / hydroxide) – low pH increases leaching, inhibits nutrient uptake and mobilises toxic aluminium and lead Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • leaching → nutrients or ions washed down out of the soil / washed away by water	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 308-310, 313; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
03.1	Marks shared Up to 6 marks in total - share them between the 4 points below according to how completely each one is made. up to 2 Organic advantages: often waste products, locally available and cheap – add humus improving structure and water retention – increase soil biota – slow release reduces leaching up to 2 Organic disadvantages: nutrient composition cannot be controlled – released slowly so needs long-term planning – bulky with high water content so expensive to transport – cannot be applied to a growing crop – odour / pathogens up to 2 Inorganic advantages: composition controlled to match crop needs – nutrients released rapidly – concentrated and easy to apply up to 2 Inorganic disadvantages: high energy (fossil fuel) manufacture (Haber process) – no organic matter so humus and biota decline – some toxic to worms – high solubility so leached causing eutrophication – raw materials (phosphate rock) finite Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • humus → dark, decomposed organic matter in soil • leaching → nutrients or ions washed down out of the soil / washed away by water • eutrophication → nutrient enrichment of water leading to algal growth Do not credit • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	6
03.2	Marks shared Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. up to 2 Crops grown in a circulating nutrient solution (inert medium) in greenhouses – nutrient concentrations monitored and adjusted so nutrients are never limiting – abiotic factors controlled up to 2 Roots are small (all in contact with nutrients) so more growth goes into the harvested crop – no soil-borne pathogens or weeds – crop stays fresh with roots up to 2 Disadvantages: high inputs of nutrients and energy (heating, lighting, pumping) – high level of technical knowledge – only economic for high-value crops Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • abiotic → non-living / physical factors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 310-312; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
04.1	Marks shared Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. up to 2 Rotation: each crop has its own pests – growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early up to 2 Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids up to 2 Rotation also evens out nutrient demands and includes legumes Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	4
04.2	1 mark each 1 mark for each of the 3 points below. 1 Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) 1 It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 1 Cultural control alone loses about 12% yield – integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. biodiversity → the variety of life: habitat, species and genetic diversity persistent → does not break down / stays in the environment for a long time Do not credit - pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
04.3	1 mark each 1 mark for each of the 3 points below. 1 Large numbers of males are sterilised with gamma radiation and released 1 Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring – enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) 1 Only the target species' females mate with them so non-target species are unaffected – fails if sterilised males behave abnormally or re-colonisation occurs Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
05.1	Marks shared Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. up to 2 Success: Cactoblastis moth introduced from South America controlled prickly pear cactus in Australia – Encarsia parasitic wasps control greenhouse whitefly up to 2 Problems: cane toads introduced to Australia to control sugar cane beetles did not control them but ate many other species and are toxic to predators – rosy wolfsnail introduced to Polynesia to control African land snails ate endemic Partulid snails (48 species extinct) – harlequin ladybirds introduced to Europe eat native ladybirds up to 2 Lesson: control agents should be specialist feeders on the pest Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • biological control → using a natural predator, parasite or pathogen of the pest • endemic → found naturally only in that one place	4
05.2	1 mark each 1 mark for each of the 2 points below. 1 Release an artificial sex scent attracting males (or females) of the pest species 1 Used to monitor when the pest arrives so pesticide is timed correctly, or (in enclosed greenhouses) to trap enough of one sex to prevent breeding Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
05.3	1 mark each 1 mark for each of the 3 points below. 1 Regular low doses kill only the most sensitive bacteria so resistant strains survive and multiply 1 Agricultural use exceeds medical use, so resistance spreads 1 Resistant zoonotic bacteria (E. coli, Salmonella, Campylobacter) can transfer to humans and are harder to treat – hence reduced use of antibiotics is a sustainability goal Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • sustainability → meeting present needs without stopping future generations meeting theirs Do not credit • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3

Question	Marking guidance	Marks
05.4	1 mark for the point below. 1 Grow nectar-rich flowering plants alongside crops – restrict pesticides harmful to bees / spray when not in flower – introduce bee hives Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 316-320; AQA spec 3.5.1.1.	